Rate blindness in archaeological science: a census of 225 papers and six case audits

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Abstract

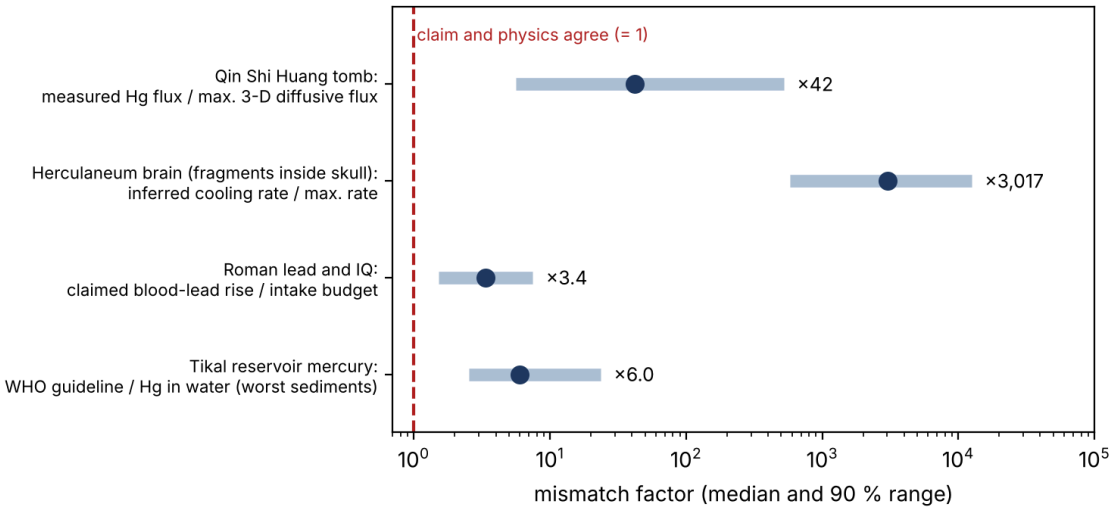
Archaeological science often moves from a measured quantity to a mechanism without checking whether the mechanism can run at the required *rate*. We call this rate blindness and measure how often it happens.

We took every open-access *Scientific Reports* paper from 2020–2025 with "archaeolog*", "ancient", "prehistoric" or "archaeometr*" in the title (225 papers). The screening protocol was pre-registered before any paper was read. For each paper whose conclusion relies on a physical, chemical or biological mechanism of checkable magnitude, we computed the relevant transport, heat, storage, intake or reaction limit from the paper's own numbers, with Monte Carlo ranges. Every case that did not pass was recalculated with independent code.

Main results:

- **21 of 225 papers (9%; 95% CI 6–14%)** rest on such a mechanism.
- Of these, **9 (43%; CI 22–66%)** do not survive the check without an assumption the paper neither states nor justifies, and **3 (14%)** fail outright. The failures are a lime "internal carbonation" that requires ~570× more CO_2 than the additive can supply, tell-erosion rates inflated ×7.6 by a unit mix-up, and drinking water deemed "seriously compromised" by sediment mercury that would yield ~11–17% of the WHO guideline.
- **Incidental numerical or unit errors** turned up in 5 of the 32 applicable or borderline papers, including modern carbon understated ×3–72 in a paleosol radiocarbon study.
- **A second pre-registered random audit (40 papers)** tested a related "scale dogma", lab-specimen results transferred to real objects without dimensional similarity. It found the problem rare (3/40 applicable, 0/3 conclusions changed). We report it as a negative result.

We complement the census with six detailed audits of widely publicised claims, including a mercury flux from the Qin Shi Huang tomb 42× above the diffusive limit, a "glass brain" cooling rate 600–12,500× faster than conduction inside a skull allows, and a Roman-lead IQ effect overstated 2–3×. Each comes with a falsifiable test.



Mismatch between claim and physical limit for the four case audits that fail (median and 90% Monte Carlo range).

1. The pattern

Archaeological science often argues from *stocks*: how much mercury is in a tomb, what glass-transition temperature a material shows, how much water a catchment provides, how much lead a sediment holds. But a mechanism also needs a *flux* on the stated time scale, and every flux has a ceiling:

- diffusion is limited by $D \cdot \Delta C \cdot S$, with S the geometric shape factor;
- conductive cooling is limited by $k/(\rho c \cdot L^2)$;
- a reservoir fed by episodic events is limited by storage between events;
- human uptake is limited by intake rates and partitioning.

The check is cheap. Identify the flux the mechanism requires, compute its ceiling with the paper's own geometry, and report the ratio.

2. Census of 225 papers

Protocol. Corpus: Europe PMC query `JOURNAL:"Sci Rep" AND (TITLE:archaeolog* OR TITLE:ancient OR TITLE:prehistoric OR TITLE:archaeometr*) AND OPEN_ACCESS:y AND PUB_YEAR:[2020 TO 2025]`, giving 225 papers.

1. **Random sample.** A sample of 40 (seed 20261101) was screened first.
2. **Census.** The remaining 185 were screened after a second registration.
3. **Classification.** A paper is *applicable* if its main conclusion relies on a mechanism acting at some magnitude or within some time, with the numbers needed for an order-of-magnitude check. Classes:
   - FALLA (fails): claim/limit ≥3 in ≥90% of Monte Carlo draws, or the main conclusion does not hold;
   - CONDICIONAL: holds only under an undeclared or unjustified assumption;
   - PASA (passes);
   - INDETERMINADO.

4. **Screening and verification.** Six independent screeners screened the papers. We recalculated every non-passing case with our own code and checked every quotation against the full text.

	k / n	% (95% CI, Clopper–Pearson)
Applicable	21 / 225	9% (6–14%)
Fail or conditional	9 / 21	43% (22–66%)
Fail	3 / 21	14% (3–36%)
Pass	10 / 21	48% (26–70%)
Sensitivity: including 11 borderline papers, fail or conditional	10 / 32	31% (16–50%)

The nine non-passing papers:

- *Fail:*
  - organic lime putty, where CO_2 from fermented extracts carbonates ≤0.2% of the lime;
  - RUSLE erosion of tells, where a US-scale K = 0.30 is combined with SI rainfall erosivity, inflating absolute rates ×7.6. The "extreme" class would imply ~20 m of lowering in 2,000 years on 23 m mounds;
  - Tikal reservoir mercury.
- *Conditional:*
  - the Tikal zeolite filter (feasible only with ponded inflow);
  - paleosol radiocarbon (DOC infiltration feasible only at the top of all parameter ranges, plus a ×3–72 unit error);
  - brown fat as the cause of Fuegian cold resistance (human-measured brown-fat capacity supplies ~1 W of the ~42 W required);
  - mm-sized metal spheres in Roman glass (colloidal agglomeration would take ~10^6 h, against firings of 3–48 h);
  - an earthquake mudflow at Lajia (needs saturated loess, not shown for 4000 BP);
  - extreme hair vitamin-D levels attributed to weight loss (hair would take ~50% of daily 25(OH)D turnover at ~1,300× serum concentration).

Previously published? We searched for prior quantitative critiques of each of the nine and found none.

Scope. The census covers one journal over six years. The rate applies to that population and is an empirical prompt, not a field-wide estimate.

Negative result. A second pre-registered audit of 40 further papers tested a "scale dogma": lab-specimen results transferred to real objects without dimensional similarity (Fourier or Damköhler numbers). Only 3 papers were applicable, and none changed its conclusion under scaling. The scale problem exists in individual cases (Section 3.2) but is not common in this corpus.

3. Case studies

These six widely publicised claims were selected *because* they looked vulnerable, before the census. Two of them (Tikal mercury and the Tikal filter) are also in the census corpus.

3.1 Qin Shi Huang: a tomb that is probably not sealed

Claim. Zhao et al. (Sci. Rep. 10, 10414, 2020) measured atomic mercury above the mound by differential-absorption lidar. They estimated an outflux of ≈5×10⁻⁸ kg s⁻¹ and wrote that mercury "may be escaping through cracks". The chamber ceiling lies ≈30 m below the surrounding ground, under a ≈50 m mound.

Test.

- Air in contact with liquid mercury saturates at C_sat ≈ 9.4 mg m⁻³ (15 °C).
- Steady diffusion through 30–80 m of compacted loess uses the Millington–Quirk diffusivity (porosity 0.30–0.45, water saturation 0.3–0.8). Lateral spreading is included through the shape factor of a disk buried in a semi-infinite medium.
- The result is a maximum flux 42× below the measured one (5.6–520). Diffusion suffices in 0.02% of 2×10⁵ draws. Typical emission from the mound soil (~190 ppb) is 9–220× too small.
- An open air path, on the other hand, could suffice. The required exchange is ~460 m³ of saturated air per day.

Caveats.

- Barometric pumping supplies this robustly only if much of the 140 × 110 × 30 m underground palace is open space.
- The excess concentration is 2.4–3.6σ per scan and was absent at the windy third site.

Predictions (pre-registered).

- Emission rises when barometric pressure falls.
- Soil-gas mercury increases with depth.
- Winter emission exceeds summer emission.
- A night-by-night ranking for the 2016 campaign, built from station pressure data.

3.2 Herculaneum: a cooling rate heat cannot deliver inside a skull

Claim. Giordano et al. (Sci. Rep. 15, 5955, 2025) measured calorimetric onsets of 420 °C at 0.42 K s⁻¹ and 510 °C at 1000 K s⁻¹ on glassy fragments "sampled intact, directly from within the skull". They inferred a natural fictive temperature T_f ≈ 510 °C and cooling "at rates comparable with our experimental 1000 K s⁻¹" after a short-lived ash cloud.

Test.

- *Cooling.* Transient conduction in a heated head, with convective and radiative loss, gives 0.1–1.7 K s⁻¹ inside the cranium.
- *Fragmentation.* The authors propose fragmentation into "sub-centimetric pieces … favouring the heat exchange". This helps only if the fragments were exposed to open air: free fragments

reach 17–720 K s^{-1}, which would imply T_f = 460–506 °C. Enclosed fragments still lose heat through the skull. The paper itself invokes the skull's protection during heating.
- *Fictive temperature.* At in-skull rates, the authors' kinetics (E = 390 kJ mol^{-1}) give T_f ≈ 405–440 °C. This overlaps the 315–465 °C burial deposits, the criterion the authors used to reject 420 °C.
- *Survival.* With the same E, the material softens in 0.4 s at 465 °C and would anneal within minutes to hours at 315–350 °C.
- *Heating.* Drying and heating the brain above 510 °C needs ≥3–40 min of full exposure.

Prior work. Morton-Hayward et al. (2020) questioned the temperatures and the survival of fatty acids in the first report; they did not address cooling rates.

Test proposed. Upscans after the first one show no events. The natural T_f should therefore be measured from first-scan onsets on fresh fragments at several heating rates, compared with reference tissue glasses of known cooling history.

3.3 Roman lead: a regression's intercept counted as pollution

Claim. McConnell et al. (PNAS 122, e2419630121, 2025) model +1.04 ng m^{-3} of air lead over Europe. They apply BLL = exp[1.932 + 0.140 ln(PbA)] from Richmond-Bryant et al. (2014; NHANES III, unadjusted). They obtain a 2.0–5.3 µg dl^{-1} "enhancement", which they add to a 1 µg dl^{-1} Neolithic baseline, and hence a deficit of 2.5–3 IQ points.

Test.

- *What the regression predicts.* It gives the total blood lead of children exposed to all 1988–94 sources: with almost clean air it already predicts 1.6 µg dl^{-1}.
- *Extrapolation.* Roman air is ~35× below the calibration median (0.036–0.037 µg m^{-3}). At 1 ng m^{-3} the implied slope is 355 µg dl^{-1} per µg m^{-3}, whereas Richmond-Bryant et al. report 15.3–16.5 at low air lead.
- *Differencing the regression.* Taking the difference against a natural background gives +1.0 µg dl^{-1} and 1.4 points, and this is still an extrapolation.
- *Intake budget.* An independent budget gives +0.71 µg dl^{-1} (0.32–1.6) and 1.0 points (0.5–1.9). It covers inhalation and deposition on cereals (Roman grain yields) and vegetables, cistern rainwater and soil, using the paper's own >500 kt deposited and child slope factors.

The effect is plausible, but 2–3× smaller than claimed for the European average.

3.4 Tikal: sediment mercury is not drinking-water mercury

Claim. Lentz et al. (Sci. Rep. 10, 10316, 2020) report sediment mercury above the "toxic effect threshold" of 1 µg g^{-1}, a criterion for benthic organisms. They conclude that the reservoirs were "seriously compromised as sources of drinking water" and that the elite were "fed foods laced with mercury at every meal".

Test.

- Mercury reaches the water dissolved (sediment–water partitioning, log K_d = 4.5–6) and on suspended particles (5–100 mg L^{-1}).
- For the most contaminated sediments (10–30 µg g^{-1}), the water carries ~1.0 µg L^{-1} (0.26–2.4). That is ~17% of the WHO guideline of 6 µg L^{-1}, which no draw exceeds.
- Intake at 3–5 L day^{-1} is ~3% of the WHO tolerable intake.
- The cyanobacteria evidence in the same paper is independent and not assessed here, nor is methylmercury in fish.

3.5 Conditional cases

Djoser (Landreau et al., PLOS ONE 19, e0306690, 2024). We ran a daily water balance with the authors' storage elements, demand, evaporation and infiltration. The lift receives 7–34% of its demand with ≤2 floods per year. It receives ~80–100% with frequent floods or near-perennial flow, which the authors propose. O. Siegel had objected qualitatively that the basin "could not have held enough water from occasional rains". Our result makes that objection quantitative and shows it depends on the flood regime. *Test:* do 3rd-Dynasty sediments in the Abusir wadi show frequent or continuous flow, or isolated floods separated by aeolian layers?

Tikal filter (Tankersley et al., Sci. Rep. 10, 18021, 2020). Passing the reservoir volume (58,000 m^3 yr^{-1}) through a slow filter needs ~550–2,300 m^2 if storm water arrives directly. It needs only ~34–200 m^2 if water ponds behind the dry-laid walls of the authors' own scheme. Seasonal drying of the biological layer remains a qualitative concern.

4. Discussion

What the census measures. Most archaeological-science papers in the corpus (91%) cannot suffer rate blindness, because they measure composition, DNA, form or age directly. Among the 9% that infer a mechanism of checkable magnitude, about four in ten (22–66%) do not survive an order-of-magnitude check made with their own numbers unless an unstated assumption is added. One in seven fails outright. The checks are cheap, typically one page of arithmetic, and none of the nine had been published.

The census is limited to one journal, and the classification involves judgement. We counter this with pre-registration, independent screening, independent recalculation and a sensitivity analysis that includes the borderline papers (31%, CI 16–50%).

Hand-picked cases. Of the six selected cases, four fail and two are conditional. Five further hand-picked claims passed or could not be decided with public data:

- the ScanPyramids Big Void;
- Teotihuacan muography;
- Mt Echia muography;
- the Hierapolis Plutonion;
- X-ray dating of the Turin Shroud.

What the census adds is the measured rate: rate checks change conclusions often enough in an ordinary, unselected corpus to be worth making routine for mechanism claims.

Our own drafts illustrate the point. An independent adversarial review of this work found:

- a one-dimensional diffusion bound that understated lateral flux by 3–6×;
- a filter target set at all runoff rather than the reservoir volume;
- a flood regime that excluded the authors' own scenario;
- a misattributed citation.

All of these are corrected here. Rate checks need the same scrutiny as the claims they test.

Data and code

All calculations, Monte Carlo inputs and figures are open:

- Qin: `simulador/qin_mercurio.py`, `simulador/qin_presion_prerregistro.py`;
- Herculaneum: `simulador/herculano_vidrio.py`;
- Djoser: `simulador/saqqara_balance_hidrico.py`;
- Roman lead: `simulador/plomo_romano.py`;
- Tikal: `simulador/tikal_filtro.py`, `simulador/tikal_mercurio.py`;
- census verification: `simulador/paleosuelo_radiocarbono.py`, `simulador/cal_organica.py`, `simulador/censo/*.py`;
- protocols, samples and results: `auditoria_aleatoria/` (PROTOCOLO.*md*, *corpus.csv*, *muestra*.csv, censo.csv, RESULTADO*.md);
- results: `resultados/42–55_*.json`.

The pre-registered predictions are P15 and P16 in `PREDICCIONES.md`.

References

Census papers that did not pass (Scientific Reports; PMCID): - Shivakumar et al. 2021, 11:13261 (PMC8225893); - Forti et al. 2023, 13:7683 (PMC10195873); - Lentz et al. 2020, 10:10316 (PMC7316703); - Tankersley et al. 2020, 10:18021 (PMC7582844); - Broz et al. 2023, 13:7476 (PMC10167333); - Watanabe et al. 2021, 11:23290 (PMC8639971); - PMC8160351 (Roman glass metal inclusions, 2021); - PMC11109126 (Lajia earthquake mudflow, 2024); - PMC11760916 (hair 25(OH)D3, 2025).

Other references: - Giordano, G. et al. Unique formation of organic glass from a human brain in the Vesuvius eruption of 79 CE. *Sci. Rep.* 15, 5955 (2025). - Landreau, X. et al. On the possible use of hydraulic force to assist with building the step pyramid of Saqqara. *PLOS ONE* 19, e0306690 (2024). - Lentz, D. L. et al. Molecular genetic and geochemical assays reveal severe contamination of drinking water reservoirs at the ancient Maya city of Tikal. *Sci. Rep.* 10, 10316 (2020). - McConnell, J. R. et al. Pan-European atmospheric lead pollution, enhanced blood lead levels, and cognitive decline from Roman-era mining and smelting. *PNAS* 122, e2419630121 (2025). - Millington, R. J. & Quirk, J. P. Permeability of porous solids. *Trans. Faraday Soc.* 57, 1200–1207 (1961). - Morton-Hayward, A. L. et al. A conscious rethink: why is brain tissue commonly preserved in the archaeological record? *STAR* 6, 87–95 (2020). - Moynihan, C. T. et al. Dependence of the fictive temperature of glass on cooling rate. *J. Am. Ceram. Soc.* 59, 12–16 (1976). - Richmond-Bryant, J. et al. The influence of declining air lead levels on blood lead–air

lead slope factors in children. *Environ. Health Perspect.* 122, 754–760 (2014). - Bower, B. Was Egypt’s first pyramid built with hydraulics? *Science News*, 5 August 2024 (O. Siegel quoted). - Tankersley, K. B. et al. Zeolite water purification at Tikal, an ancient Maya city in Guatemala. *Sci. Rep.* 10, 18021 (2020). - WHO. Mercury in drinking-water (chemical fact sheet, 2022). - Zhao, G. et al. Mercury as a geophysical tracer gas — emissions from the emperor Qin tomb in Xi’an studied by laser radar. *Sci. Rep.* 10, 10414 (2020).